Volume of pyramids and cones

1

a $160\ \text{cm}^3$ b $200\ \text{cm}^3$ c $154.78\ \text{cm}^3$ d $100\ \text{cm}^3$

e $282.33\ \text{mm}^3$

2 $\frac{448}{3}\ \text{cm}^3$

3 $2\,561\,328\ \text{m}^3$

4 12 cm

5

a i 128.45 m ii $85\,633.33\ \text{m}^3$

b i 13.92 cm ii $125.3\ \text{cm}^3$

6 $2\,574\,466.7\ \text{m}^3$

7

a $4608\ \text{mm}^3$ b $3969\ \text{mm}^3$

8 26 mm

9 18 cm

10

a $25.13\ \text{cm}^3$ b $32.45\ \text{cm}^3$ c $56.55\ \text{units}^3$

11

a $220.0\ \text{cm}^3$ b $452.4\ \text{cm}^3$ c $1187.5\ \text{cm}^3$

12

a $43.37\ \text{m}^3$ b $36.65\ \text{cm}^3$

13

a $871.27\ \text{cm}^3$ b $103\,823.35\ \text{m}^3$

14

a 16.0 m

b 25 cm

15

a 19.9 cm

b 8 mm

c 5.72 mm

16 $h = \sqrt[3]{H^3 \dfrac{x}{V}}$

Additional questions

17

a 440 cm^3

b 24.44 in^3

c 84.82 cm^3

d 1526.81 km^3

18 55 in^3

19 6 units

20 45 m^2

21 Yes, the silo has a height of 10 ft which is shorter than the 12 ft clearance of the doorway.

22 You can order either 3 of Option 1 or 1 of Option 2 to maximize the amount of kettlecorn you buy. Since the cone and the cylinder have the same diameter and height, three of the cones is equal to the volume of one cylinder. Both orders will cost $\$10.50$ total.